

Courtney A. Jimenez

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RESEARCH PROFILE

Computational social neuroscientist specializing in the neural and cognitive mechanisms of social learning, memory, and person knowledge. Extensive experience with high-dimensional fMRI data analysis, reinstatement-based memory consolidation paradigms, and computational modeling of human social cognition using reinforcement learning. Seeking a Data Scientist role to bridge high-dimensional data analysis with computational modeling of human social behavior.

EXPERIENCE

Columbia University — Columbia Social Neuroscience Lab

New York, NY

Ph.D. Candidate, Columbia Psychology Department (PI: Meghan L. Meyer)

Sep. 2023–May 2026

- Designed and conducted computational modeling studies investigating how person knowledge learning biases social memory and subjective connection using reinforcement learning models of human social cognition
- Applied semantic similarity analysis using sentence-transformers (S-BERT) to quantify linguistic patterns in free recall data, bridging NLP methods with social cognitive research
- Completed Computational Social Cognition Summer School (University of Birmingham, 2025) — 2-week intensive training in computational modeling methods (Bayesian hierarchical modelling, sequential-sampling methods, RL)
- Presented work at CNS, SANS, and OHBM annual meetings; received multiple travel awards for conference presentations
- Mentored undergraduate and junior graduate researchers; TA for 4 courses including Research Methods: Human Behavior, Moral Psychology, and Abnormal Psychology

Dartmouth College — Dartmouth Social Neuroscience Lab

Hanover, NH

Ph.D. Student, Psychological and Brain Sciences (PI: Meghan L. Meyer)

Sep. 2019–Aug. 2023

- Designed and conducted fMRI studies examining how the dorsomedial prefrontal cortex (DMPFC) prioritizes social information consolidation during rest; findings published in PNAS (2024)
- Built end-to-end fMRI preprocessing and analysis pipelines (FSL, Nilearn, SPM) including GLM, ROI, and neural reinstatement analyses on HPC clusters (SLURM) in containerized environments (Singularity, Docker, Conda)
- Collected and analyzed data (R) on changes in social working memory across adolescent development
- Completed Neurohackademy (Univ. of Washington, 2023) — 2-week intensive training in open neuroimaging data science (Python, Reproducibility, Git/GitHub, Docker, Machine Learning, Deep Learning)
- TA for 4 courses including Laboratory in Psychological Science, Introductory Psychology, and Social Psychology

University of California, Davis — Center for Mind and Brain

Davis, CA

Jr. Specialist, Brain and Social Cognition Lab (PI: Lindsay Bowman, Ph.D.)

Jul. 2017–Aug. 2019

- Lead research analyst on project investigating the relationship between mother-infant interaction and infant brain electrophysiological responses using power spectral density analysis and neural time-series analysis
- Developed Matlab scripts integrating EEGLab, ERPLab, and EP Toolkit for electrophysiological data processing and subsequent neuroimaging analysis
- Facilitated and collected EEG data for >50 children ages 3-5 years old to investigate the neural correlates of social cognitive development, including theory of mind and emotional responses to facial stimuli
- Developed experimental stimuli in E-Prime integrated with BrainVision, Tobii, and OptiTrack systems for data collection and trained undergraduate students in EEG data collection and analysis methods

Jr. Specialist, Cognitive Electrophysiology and Neuroimaging Lab (PI: John Olichney, M.D.)

May 2017–Aug. 2019

- Facilitated ongoing research investigating the clinical utility of ERPs as a diagnostic tool in tracking the onset and progression of various neurodegenerative disorders
- Wrote and maintained Linux Bash scripts for EEG/ERP data processing and analysis (BASH, DOS, Ruby)
- Scored neuropsychological exams such as ADAS-COG, CVLT, MMSE, MOCA, PACC
- Maintained databases and IRB compliance for longitudinal NIH R01 funded research
- Trained research assistants on EEG data processing in Linux and DOS environments

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EDUCATION

Columbia University

Ph.D. in Psychology

M.Phil. in Psychology

New York, NY

May 2026

May 2025

Dartmouth College

M.A. in Psychological and Brain Sciences

Hanover, NH

June 2023

University of California, Davis

B.S. in Neurobiology, Physiology, and Behavior

B.A. in Psychology

Davis, CA

Mar. 2017

Mar. 2017

PUBLICATIONS

Jimenez, C.A., & Meyer, M. L. (in preparation). Reinforcement learning underlies social connection and memory in a friendship decision-making task.

Jimenez, C. A., & Meyer, M. L. (2024). The dorsomedial prefrontal cortex prioritizes social learning during rest. *Proceedings of the National Academy of Sciences*, 121(12), e2309232121.

Yeum D, **Jimenez C.A.**, Emond J.A., et al. (2023). Differential neural reward reactivity in response to food advertising medium in children. *Frontiers in Neuroscience*.

Jimenez, C.A.*, Geisler, D.*, Smith, M., & Meyer, M. L. (in preparation). Changes in Social Working Memory Across Adolescent Development.

TECHNICAL SKILLS

Programming: Python, R, MATLAB, Bash/Unix, Git/GitHub, Conda/Docker, Jupyter Notebook, R Markdown, SLURM/HPC

Neuroimaging: fMRI analysis (GLM, ROI, whole-brain, reinstatement/neural pattern similarity), NITools, Nilearn, NiBabel, SPM, BIDS formatting, fmriPrep, MRIQC

Statistics & Quantitative Methods: Inferential statistics (ANOVA, MANOVA, t-tests, chi-square, non-parametric equivalents), mixed-effects & multilevel models (lme4, nlme), multiple comparisons correction (FDR, Bonferroni), contrasts & planned comparisons, bootstrap & permutation resampling

Computational Modeling: Reinforcement learning implementation & fitting (Rescorla-Wagner), parameter & model recovery, model comparison (BIC/AIC/WAIC)

Machine Learning: scikit-learn, sentence-transformers (S-BERT), supervised & unsupervised learning, dimensionality reduction (PCA, ICA), deep learning fundamentals (CNNs, RNNs, Transformers), semantic similarity analysis, NLP (NLTK)

Research & Experimental Methods: Experimental design, PsychoPy, Qualtrics, Prolific/MTurk, mixed methods

Visualization: Matplotlib, Seaborn, ggplot2, brain surface visualization (MNI space), publication-quality figures

HONORS & AWARDS

Psychology Graduate Travel Award

Mar. 2026

SANS Innovation Award Finalist

Apr. 2025

ASGC/GSAS Travel Award

May 2024

Columbia GSAS Provost Diversity Fellow

2023–2024

Neukom Institute for Computational Science Travel Grant

Apr. 2022

Dartmouth E.E. Just Liftoff Fellow

2019–2020

TALKS

Blitz Talk, Social and Affective Neuroscience Society, Toronto, CA

Apr. 2024

Zuckerman Institute Columbia Neuroscience Seminar Lab Meeting

Mar. 2024

Data Blitz, Cognitive Neuroscience Society, San Francisco, CA

Apr. 2023

WORKSHOPS

Computational Social Cognition Summer School, Birmingham, UK

September 2025

Neurohackademy, University of Washington, Seattle, WA

August 2023